

SEC 2-FA2 GROUP 3-CELLONA, BD; RICAFORT, NF-FA2

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2. An experiment consists of tossing two fair coins. Use R to simulate this experiment 100 times and obtain the relative frequency of each possible outcome. Hence, estimate the probability of getting one head and one tail in any order.

```
# Toss two coins 100 times
coin1 <- sample(c("H", "T"), 100, replace = TRUE)
coin2 <- sample(c("H", "T"), 100, replace = TRUE)

# Combine the outcomes
outcomes <- paste(coin1, coin2, sep = "")

# Frequency of each outcome
table(outcomes)
```

```
## outcomes
## HH HT TH TT
## 23 27 24 26
```

Relative Frequencies

```
# Find the relative frequency
table(outcomes) / 100
```

```
## outcomes
##   HH   HT   TH   TT
## 0.23 0.27 0.24 0.26
```

Probability of One Head and One Tail

One head and one tail can occur as **HT** or **TH**.

```
# Estimate the probability
probability <- sum(outcomes == "HT" | outcomes == "TH") / 100

probability
```

```
## [1] 0.51
```

The estimated probability of getting one head and one tail is given by the simulation result above. It should be close to **0.5**.